

SIMSON M

Software Development Engineer II

Java • Spring Boot • Microservices • React • AI Systems

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PROFESSIONAL SUMMARY

Software Development Engineer with **3 years of experience** designing and delivering enterprise and AI-powered applications using **Java, Spring Boot, microservices, React, and distributed systems**. Experienced in building secure backend services, REST and gRPC APIs, event-driven architectures, RBAC, caching, authentication, and enterprise workflow platforms across **banking, aviation, SaaS, and conversational AI**. Currently working on **AI voice-bot evaluation and conversational AI systems**, with experience across the complete software lifecycle from system design and backend development to frontend integration and cloud deployment.

TECHNICAL SKILLS

Languages	Java 8, Java 17, TypeScript, JavaScript, SQL
Backend	Spring Boot, Spring Security, Microservices, REST APIs, gRPC, Protocol Buffers
Frontend	React.js, Angular, TypeScript, Tailwind CSS, Redux, Context API
Databases & Cache	PostgreSQL, MySQL, MongoDB, Microsoft SQL Server, Redis
Messaging	Apache Kafka, Event-Driven Architecture
Security	Keycloak, OAuth2, JWT, RBAC, Azure AD, Custom Keycloak SPI
AI & Voice	Voice Bot Evaluation, LLM APIs, Prompt Engineering, Structured Evaluation, Token Usage Analysis, WebSocket Audio
Cloud & DevOps	AWS, Azure, Docker, Kubernetes, Azure DevOps, CI/CD, Git

PROFESSIONAL EXPERIENCE

KAPTURE CX

Bengaluru, India

Software Development Engineer II

Jul 2026 – Present

AI Voice Bot Evaluator — Conversational AI Testing Platform

Software Development Engineer II

AI Voice Bots · LLMs · WebSocket · Automated Evaluation · Analytics

- Designed and developed an **AI Voice Bot Evaluation platform** for systematically testing conversational agents across functional behaviour, conversation quality, expected responses, prohibited behaviours, and scenario-specific objectives.
- Built a configurable **automated test-scenario framework** supporting persona, language, objective, expected behaviours, forbidden behaviours, severity, maximum conversation turns, and structured evaluation criteria.
- Developed evaluation workflows for **live customer conversations and voice-agent test sessions**, enabling completed and active test cases to be tracked throughout an evaluation run.
- Implemented **token-usage analysis** to track conversational AI consumption at conversation and evaluation level, enabling detection of unnecessarily expensive or verbose bot interactions and supporting optimisation recommendations.
- Worked on **Translation Intelligence and Translation Gateway** capabilities for evaluating multilingual voice-bot conversations and translation behaviour across supported languages.
- Designed integration approaches for existing voice bots using **WebSocket-based audio streaming**, conversation ingestion, transcripts, audio recordings, and knowledge-base documents for post-conversation evaluation.

- Developed structured evaluation reports that surface test-case results, conversation-level observations, quality issues, and actionable recommendations for improving voice-agent behaviour.

MINDGRAPH TECHNOLOGIES

Chennai, India

Full Stack Developer

Nov 2023 – Jul 2026

OIS-KLIA — Malaysia Airports Holdings Berhad (MAHB)

Full Stack Developer

Java · Spring Boot · React · MongoDB · Redis · Azure AD

- Designed and developed **Spring Boot services** supporting multi-stage operational workflows across multiple business processes for Malaysia Airports.
- Designed the MongoDB persistence layer with **UTC-based storage and Malaysia-time-aware retrieval**, providing consistent date and time handling across distributed application workflows.
- Engineered an integration layer to **synchronise user and organisational information from Active Directory**, including user profiles, departments, groups, roles, and reporting relationships.
- Implemented **Redis caching and fine-grained RBAC** to reduce repeated database access while enforcing module and permission-level access control.
- Built secure **React–Spring Boot REST integrations** with role-aware UI rendering and backend authentication for enterprise application modules.
- Integrated **SendGrid-based event notifications** for workflow events, approvals, exceptions, and operational communications.

AWF — Anti-Money Laundering Workflow System (UOB)

AppWorks Developer

OpenText AppWorks · BPM · Workflow Automation · RBAC

- Developed enterprise **AML investigation and escalation workflows** for United Overseas Bank using the OpenText AppWorks platform.
- Designed configurable **multi-stage BPM workflows** covering case initiation, risk assessment, compliance review, escalation, and audit closure.
- Implemented configurable business rules and conditional routing for complex AML decision flows while maintaining auditable lifecycle state transitions.

SEAS DIGILEAP — Digital Platform

Backend Developer

Java · Spring Boot · gRPC · Keycloak · Protocol Buffers

- Designed and developed a **four-service microservices architecture** covering user, profile, media, and digital-management domains using Spring Boot.
- Implemented **gRPC and Protocol Buffers** for typed inter-service communication between distributed backend services.
- Developed a **custom Keycloak SPI** for tenant-aware user onboarding and authentication, supporting multiple client applications through centralised identity and access management.
- Implemented OAuth2/JWT-based API security and role mapping for protected microservice endpoints.

Additional Enterprise Projects — MAHB

Backend Developer

Java · Spring Boot · Redis · RBAC · REST APIs

- Developed backend capabilities for a **Contract Lifecycle Management platform**, implementing multi-stage approval workflows, dynamic permissions, and Redis-backed data access.
- Built document-management workflows supporting **Creator, Reviewer, and Watcher roles**, parallel review processing, controlled document access, and automated event notifications.

PROJECTS

HASKELAI LMS — Learning Management System

Full Stack Developer

Java · Spring Boot · Kafka · Redis · MySQL · React · Docker

- Designed and developed a **microservices-based Learning Management System** supporting learner, organisation, course, and administration workflows.
- Implemented course delivery, learner progress tracking, **role-based access control**, and administrative capabilities across Spring Boot services and a React frontend.
- Integrated **Kafka** for asynchronous event processing, **Redis** for caching, and Amazon S3-compatible object storage for learning media and documents.
- Containerised application services using **Docker** and configured reverse-proxy-based deployment for independently deployable frontend and backend services.

EDUCATION

BACHELOR OF ENGINEERING — COMPUTER SCIENCE

Karpagam College of Engineering